

PS26234 — Smart Food Waste Reduction & Redistribution

SIH 2026 Prototype Plan: Features, Tech Stack & Ready-Made Datasets

Software Edition · Agriculture, FoodTech & Rural Development · Ministry of Food Processing Industries (MoFPI) · Prepared for Anuj Verma

What This Problem Statement Actually Wants

In simple words: the government wants a system that helps kitchens, hostels, and food factories waste less food. It should guess how much food will be needed (so less is over-made), spot food that is going bad before it's thrown away, send extra food to NGOs or shelters instead of the bin, and show reports on how much waste and money was saved.

The official write-up also mentions IoT sensors, large-scale logistics networks, and full ESG compliance reporting — that's too much to build in a hackathon. The plan below covers every part of the problem statement in a scaled-down, realistic way that you can actually finish and demo.

Why This Version Stands Out

A few strategic additions on top of the core build turn this from “a working prototype” into “a prototype with a clear edge” — without changing the tech stack.

- **Prevents waste, not just reports it** — the predictor suggests a same-day fix, not only a warning.
- **A two-way marketplace, not a one-way alert** — NGOs and vendors can “claim” surplus, and there's a cost-recovery channel alongside the charity one.
- **Covers food processing units too** — most teams only build the canteen side because it's easier to demo; this plan covers the half of the PS they'll likely skip.

Core Features to Build (Hackathon MVP)

1. Demand & Surplus Predictor

Staff enter how many meals were made and eaten on past days (or you load the ready dataset below). The system predicts tomorrow's need and flags likely surplus. **Unique twist:** instead of only warning about surplus, it suggests a same-day fix — e.g. “yesterday's extra rice → today's pulao special” — so it prevents waste, not just reports it.

Covers: “predict food demand in real time.”

2. Food Freshness Checker (Photo Upload)

Upload a photo of stored food or produce. The AI tags it Fresh / Near Expiry / Spoiled.

Covers: “identify food nearing expiry using image-based quality assessment.”

3. Two-Way Surplus Routing (NGO Donation + Discounted Resale)

Flagged food is routed down one of two channels: near-expiry or spoiled-soon food goes to the nearest NGO or shelter as a donation; still-good surplus goes to a discounted-resale list for nearby vendors or staff. Either side can tap “Claim” (a WhatsApp reply or an in-app button), which updates the dashboard live — a real two-way match instead of a one-way alert.

Covers: “connect surplus food with NGOs/food banks/secondary buyers through an automated redistribution network” — and gives the institution a cost-recovery reason to adopt it, not just a CSR one.

4. Pickup Route Suggestion

Shows the best route from the kitchen to whichever NGO or vendor claimed the food, with an estimated pickup time.

Covers: “optimise transportation using AI-based logistics planning” (simplified for a demo).

5. Impact Dashboard

Simple charts: meals saved, kilograms of waste avoided, estimated CO2 saved, and money saved — now also tracking money recovered from the resale channel alongside meals saved through donations.

Covers: “sustainability analytics, carbon footprint reduction, ESG reporting.”

6. Processing-Unit Waste Log (Lightweight)

A simple form/table where a food processing unit logs raw-material losses, machine downtime, or overproduction per batch. It reuses the same predictor and dashboard, so it's a small add-on rather than a second app.

Covers the “food processing units” half of the problem statement that most teams skip because the kitchen side is easier to demo.

Bonus Features (Only If Time Remains)

- A QR code at the kitchen counter so staff can log a food batch in one scan
- A leaderboard ranking kitchens or hostels by waste reduced (encourages adoption)
- A one-click “Download Impact Report” (PDF) button for judges

Recommended Tech Stack

Picked to be quick to build and easy to connect together — nothing here needs heavy setup.

Layer	Suggested Tool	Why
Frontend (web dashboard)	React.js	Clean dashboards, plenty of ready-made chart components
Mobile option (optional)	Flutter	Only needed if kitchen staff want an app on their own phone
Backend	Python + Flask	Easiest to connect with AI models; fast to build under time pressure
Database	Firebase (Firestore)	No server setup needed — big time-saver for a hackathon
AI — Freshness detection	TensorFlow/Keras + MobileNetV2 (pretrained)	You fine-tune a ready model on the dataset below instead of training from zero
AI — Demand prediction	scikit-learn (Random Forest)	A few lines of code on the ready dataset below; no deep learning needed
Maps & routing	Google Maps Directions API	Free tier is enough for a demo; gives route + time instantly

Layer	Suggested Tool	Why
Alerts & claim replies	Twilio WhatsApp API	Sends the surplus alert and, via a webhook, captures the NGO/vendor's "Claim" reply — free sandbox mode is fine for a demo
Hosting	Netlify (frontend) + Render (backend)	Free, fast to deploy, reliable for a live judge demo

Ready-to-Use Datasets (No Data Collection Needed)

You don't need to design a schema or collect your own data for this prototype — these are ready, labelled, downloadable files. Just download the folder or CSV and point your code at it.

Feature	Dataset	Where to Get It	What's Inside
Demand Predictor	Food Demand Forecasting Dataset (Genpact / Analytics Vidhya Hackathon, 2018)	Search "Food Demand Forecasting" on Kaggle (several mirrors exist, e.g. under kannanaikkal, sureshmecad)	145 weeks of real meal orders across cities and kitchens in one ready train.csv file
Freshness Checker	Vegetable Image Dataset	kaggle.com/datasets/misrakahmed/vegetable-image-dataset	~21,000 photos of 15 vegetables, already sorted into train / validation / test folders
Freshness Checker (bigger option)	FoodClass Dataset	data.mendeley.com/dataset/s/5fjkkv5xgt/2	85,000+ photos of fruit, vegetables, meat and fish, each already labelled Fresh or Rotten

How the Pieces Connect

- 1 Kitchen or processing-unit staff log today's food count or batch data (or the app pulls in yesterday's leftover number).
- 2 The Demand Predictor shows tomorrow's expected need, a surplus warning, and a same-day menu fix suggestion.
- 3 If a batch is flagged Near Expiry or Spoiled, the Freshness Checker confirms it from a photo.
- 4 Flagged food is routed: near-expiry → nearest NGO for donation; still-good surplus → discounted resale list. Either side taps Claim.
- 5 The Dashboard updates with meals saved, kilograms saved, CO2 saved, and money recovered.

Suggested Day-by-Day Build Plan

Stretch or compress this to however many days you actually have before submission.

Day	Focus
Day 1	Set up Firebase project, basic React dashboard skeleton, load the demand dataset
Day 2	Build the demand prediction model (scikit-learn) and connect it to dashboard charts

Day	Focus
Day 3	Build the Freshness Checker (fine-tune MobileNetV2 on the vegetable dataset) and the photo-upload screen
Day 4	Build the two-channel routing (NGO donation + discounted resale), the Claim flow (WhatsApp webhook or button), and the Google Maps route screen, using a small mock NGO/vendor list
Day 5	Add the lightweight processing-unit waste log, polish the dashboard, add impact numbers, rehearse the demo

Quick Tips for the Demo

- Judges care more about a working demo than a long feature list — get the 6 core features solid before touching bonus ones.
- Use 2–3 sample kitchens and a handful of mock NGOs/vendors; you don't need real partnerships for a prototype.
- Show one live example end-to-end (surplus flagged → routed → claimed → dashboard updates) — it tells the whole story in under two minutes.
- Anchor the demo to one real, specific institution (e.g. your own college mess or a nearby hostel) instead of a generic pitch — judges respond better to a grounded example than an abstract platform.